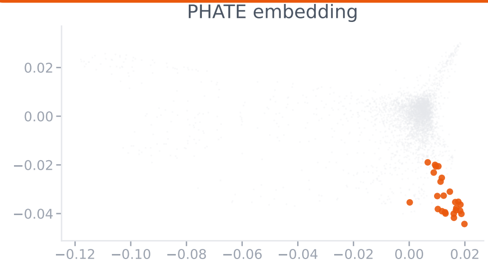
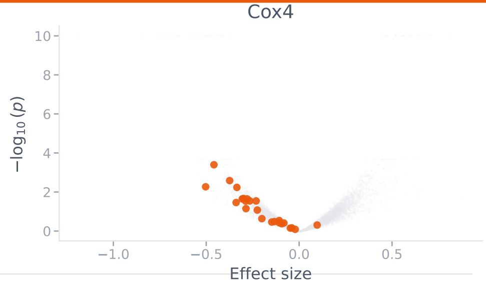
Ubiquitin-proteasome system (UPS) and protein degradation

Cluster 6





Summary

Cluster 6 shows HIGH confidence for the ubiquitin-proteasome system (UPS) pathway, with 68% (17/25) of genes being established proteasome components. The cluster contains 11 core 20S proteasome subunits (PSMA1/4/7, PSMB1/3/4/7), 6 regulatory 19S ATPase subunits (PSMC1-6), 2 non-ATPase regulatory subunits (PSMD2/7), and key neddylation machinery (NEDD8, NAE1) that regulates cullin-RING E3 ligases. ...

Established genes

`PSMA1, PSMA4, PSMA7, PSMB1, PSMB3, PSMB4, PSMB7, PSMC1, PSMC2, PSMC3, PSMC4, PSMC5, PSMC6, PSMD2, PSMD7, NEDD8, NAE1`

Novel & uncharacterized genes

FNBP4 *novel*

No functional annotation provided. Gene name suggests formin-binding protein function (typically cytoskeletal), making co-clustering with proteasome components unexpected. However, without

FOXN3 *novel*

Transcriptional repressor involved in DNA damage-induced cell cycle checkpoints. May regulate expression of proteasome components or UPS pathway genes in response to cellular stress, though this

SNRNP27 *novel*

Putative mRNA splicing factor with minimal characterization. Co-clustering with proteasome suggests potential novel link between splicing and protein degradation pathways, similar to SNRPE but with

NEDD8-MDP1 *unc.*

Gene name suggests NEDD8-related function but lacks functional annotation in provided data. Given NEDD8's established role in neddylation and cullin regulation within UPS, this gene may represent an

SNRPE *novel*

Well-characterized spliceosomal component with no established connection to proteasome function. Co-clustering is surprising and suggests potential novel crosstalk between splicing machinery and UPS.

AKIRIN2 *novel*

Recently discovered to bridge proteasomes and nuclear import receptor IPO9, mediating nuclear proteasome import. While this function is now established (2021 publication), its co-clustering with

ENSG00000256206 *unc.*

Completely uncharacterized gene with no functional annotation. Clusters with core proteasome components, suggesting potential role in protein degradation or proteasome assembly/regulation. High

ENSG00000256646 *unc.*

Completely uncharacterized gene with no functional annotation. Co-clustering with proteasome machinery suggests involvement in UPS pathway. Represents high-priority target for novel proteasome-

17 established · 5 novel · 3 uncharacterized